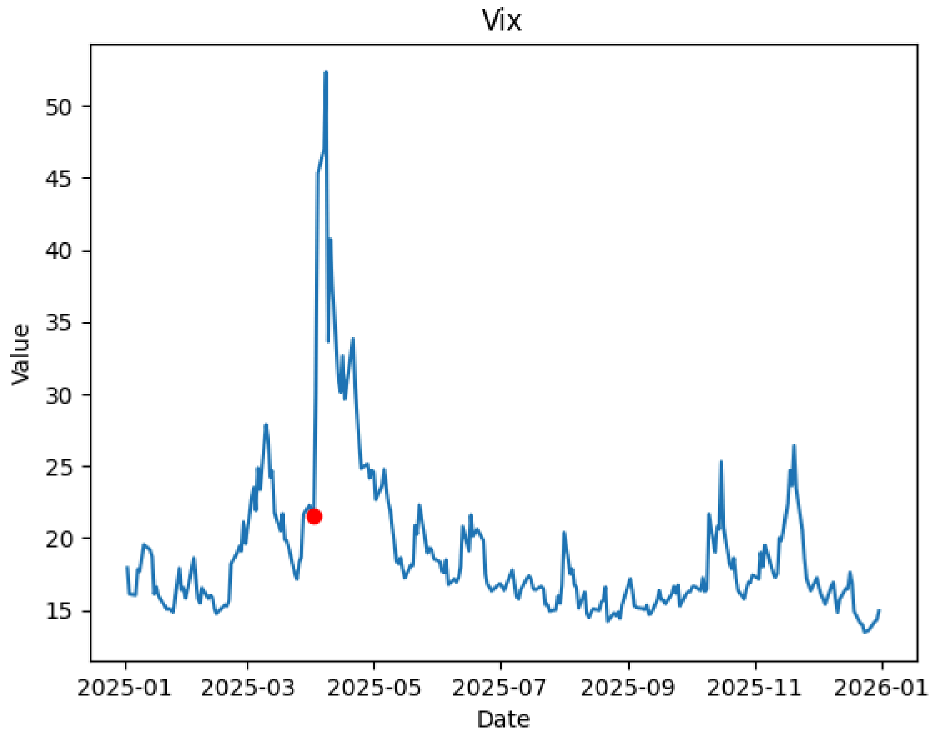
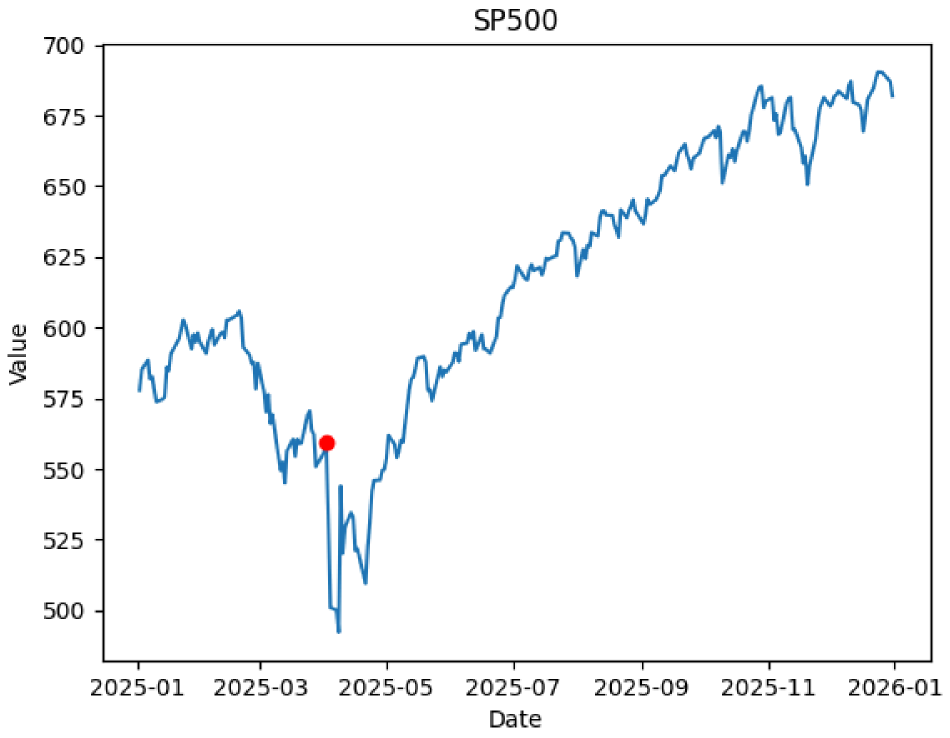
Managing Volatility Risk

All financial institutions are structurally exposed to volatility through their positions in risky assets (vega exposure).
The year 2025 has reminded us that this exposure can become critical, with major volatility spikes triggered by exogenous shocks.
The announcement of tariffs by Donald Trump caused a sudden market dislocation, visible simultaneously on the VIX and the S&P 500.
In this context, the question is not to anticipate volatility, but to know how to protect against it effectively.





We can see that, in the days following the announcement of tariffs (2025-04-02: red dot), there was a spike in the VIX and a sharp drop in the S&P 500. The metrics are evaluated from 2025-03-24 to 2025-04-11 (20 days), except for the correlation (on 2025).

VIX flight shock
+ 205 %

MDD
-35.75%

Correlations :
-0.86

SPY shock
amplitude: -13 %

MDD
-13.72%

One can be exposed to volatility in several ways: directly through futures or ETNs on the VIX, or via index options (buying puts on the S&P 500) to combine directional protection and volatility exposure. Long vega strategies (straddles, strangles) capture increases in volatility, regardless of market direction. Finally, variance swaps allow the transfer of realized volatility risk without directional exposure.

We have mentioned various products to hedge against market volatility. We will now focus on the variance swap.

Swap Variance:

- Allows an investor to hedge against an increase or decrease in the implied volatility of an underlying asset (regardless of the underlying's direction).
- It is an OTC contract between two parties:
  - Payer of fixed variance (strike set at inception)
  - Payer of realized variance (calculated over the contract

period)

- $payoff = N_{Var}(Var_{réalisé} - Var_{Strike})$

Example :

At the start of Donald Trump's term, an uncertain political context generated high market volatility. We decide to hedge using a variance swap, taking a long vega position. Assuming that we have:

- Exposure: $1,000 per 1% of volatility
- Variance strike: 20%
- Expected change in volatility: Δσ = 1%

The notional required to hedge our vega is calculated as follows:

$$N_{\text{var}} = \frac{\text{Vega exposure}}{2 \cdot \sigma_{\text{strike}} \cdot \Delta\sigma} = \frac{1\,000}{2 \times 0.20 \times 0.01} = 250,000$$

Thus, in the event of a 1% increase in volatility, the swap payoff exactly offsets our exposure.